



Project Title:

Smart Healthy Restaurant Management System

نظام إدارة المطاعم الصحي الذكي

Abstract:

In recent years, awareness of healthy eating habits has significantly increased, leading many individuals to seek better control over their daily calorie intake. However, most traditional restaurant systems focus only on order management and pricing without providing any nutritional insights about the meals offered. This creates a gap between customer needs and the services provided by restaurants, especially for individuals following diet plans or aiming to maintain a balanced lifestyle.

This project presents a Smart Healthy Restaurant Management System developed using Object-Oriented Programming (OOP) concepts in C#. The system is designed to enhance the traditional restaurant experience by integrating calorie calculation and health classification into the ordering process. Users can browse a digital menu, select meals, and instantly view their total calorie intake along with a classification of the meal as healthy, moderate, or high in calories.

In addition, the system automates order processing, calculates total prices accurately, and stores order history for future analysis. By combining efficient restaurant management with health-focused features, the proposed system aims to improve customer decision-making, reduce human errors, and promote healthier eating habits in a simple and user-friendly environment.

Problem Definition:

Many traditional restaurants rely on manual systems or basic digital solutions that only handle orders and billing. These systems do not provide any information about the nutritional value of meals, which makes it difficult for customers to make informed food choices.

Customers who are concerned about their health or following specific diets often struggle to estimate calorie intake when eating out. At the same time, restaurant owners lack tools to analyze food consumption patterns or offer healthier alternatives.

This results in poor decision-making, lack of health awareness, and inefficient restaurant management processes.

Recommended Solution:

To address these issues, we propose developing a Smart Healthy Restaurant Management System using C# and OOP principles.

The system will provide:

- A digital menu with detailed meal information
- Automatic calorie calculation for selected items
- Classification of meals into healthy, moderate, or high-calorie categories
- Automatic price calculation and bill generation
- Order management system for handling multiple customers

The system will be designed using Object-Oriented Programming concepts such as classes, inheritance, encapsulation, and polymorphism to ensure modularity, scalability, and maintainability.

Objectives:

- Develop a smart and user-friendly restaurant system
- Help customers make healthier food choices
- Provide accurate calorie tracking for meals
- Reduce errors in order processing and billing
- Apply OOP concepts in a real-world project
- Improve restaurant efficiency and service quality

Team Members:

Student ID	Student Name	Section No
1000324251	Youssef Mohamed Mahmoud Mohamed Roshdy	48
1000324138	Youssef Mohamed Mahmoud Mohamed Allam	48
1000324687	Youssef Mohamed Kamal Abdelmegid Ashour	48
1000324598	Youssef Mohamed Youssef Mohamed Youssef	48
1000323834	Youssef Mamdouh Mohamed Youssef Ali	48
1000323999	Youssef Walid Ebrahim Abdo Elkhayat	48